Pranava Karthekeyan Ponvinayagan

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Education

- **Georgia Institute of Technology** — Atlanta, GA
  B.S. Electrical Engineering (Circuit Technology & Bioengineering) | GPA: 4.0 — *Expected May 2029*

Experience

- **Neuroloops @ Georgia Institute of Technology** — Atlanta, GA
  Undergraduate Researcher — *Sept. 2026 – Present*
  - **Neuromorphic Tactile Restoration Wearable**: Engineering a closed-loop hybrid BCI wearable that restores the sense of touch by acquiring EEG & EMG signals, decoding them with a spiking neural network, and modulating TENS stimulation of skin mechanoreceptors, activating peripheral nerves that signal the CNS to evoke tactile sensation
    - Designed the full hardware system end-to-end (part selection, schematics, 4-layer flex PCB layout with stiffened component islands) as a modular wearable made up of a galvanically isolated stimulation hub, interchangeable 8-channel ADS1299 EEG/EMG modules (65×38 mm each, two clock-synchronized for 16 channels), and custom flex FPC interconnects
    - Built a ∼90V biphasic TENS stimulator (boost converter and H-bridge with hardware shoot-through interlock & current limiting), protected by a hardware fault latch that cuts stimulation on overcurrent, out-of-range electrode impedance, over-temperature, or supply brownout, and added stimulation-triggered analog-switch blanking that protects the EEG/EMG front ends independent of firmware
    - Developing embedded firmware for synchronized 16-channel EEG/EMG acquisition & real-time signal processing, isolated SPI/I2C control of the stimulation hub, electrode-impedance calculation from live current/voltage sensing, impedance-adaptive biphasic TENS waveform generation, and fault monitoring & recovery
    - Working toward porting the SNN onto a custom FPGA neuromorphic processor (board bring-up & RTL) to decode live EEG/EMG features and drive stimulation timing in real time

- **Movement Neuroscience Lab @ Northeastern University** — Boston, MA
  Visiting Researcher — *June 2026 – Aug. 2026*
  - **EEG-Modulated Neurostimulation Platform for Force Tracking**: Orchestrated a full-stack, closed-loop neurostimulation platform unifying multi-modal brain stimulation, visuomotor force-tracking, and EEG-feature-based modulation on a synchronized timeline, enabling novel paradigms such as a subject tracing the stimulation waveform being delivered to their own brain in real time
    - Architected 5 real-time C++ modules using the RTXI framework, including a 6-input force collector/processor, a 32-channel EEG collector (via LSL) with real-time spectral feature extraction from a Brain Products ActiChamp amplifier, a unified stimulation engine driving 8 configurable channels (tES, dual TI channels, a sum-of-sines channel, and TMS triggering), a fully designable per-trial task generator, and a physics-calibrated subject-facing force display
    - Integrated real-time closed-loop control that adjusts stimulation amplitude from live, baseline-calibrated EEG band power with slew-rate limiting, eliminating manual amplitude adjustment and enforcing hardware safety
    - Developed a Python orchestrator that automates full experiment deployment, including RTXI environment build and launch, module parameter configuration, live waveform preview, a preset system, and unified hardware I/O routing, turning a manual multi-step setup into a single reproducible workflow for lab-wide use

  Research Intern — *April 2026 – May 2026*
  - **Real-Time TMS-EMG Acquisition and Control System**: Developed C++ RTXI modules on Xenomai real-time Linux, replacing a non-real-time setup with programmatic TMS control and multichannel EMG acquisition for live human MEP experiments
    - Designed and hardware-validated a serial communication plugin for the Magstim BiStim^2 paired-pulse TMS device around a state-machine architecture with a checksummed command protocol, confirming reliability via loopback testing and delivering real-time, closed-loop-ready stimulation control
    - Built a 16-channel EMG acquisition module for the Delsys Trigno system using a dual-socket TCP architecture and a wait-free, non-real-time listener thread, enabling lag-free data streaming into the RTXI pipeline

Skills

- **Lab & Bench:** SMD Soldering, PCB Assembly & Bring-up, Hardware Validation & Debugging, 3D Printing, Test Equipment (Oscilloscope, DMM), Biosignal Instrumentation (EEG, EMG, TMS)
- **Hardware Design & CAD:** Schematic Capture & PCB Layout (KiCad, EasyEDA), Flex PCB Design, Analog Front-End Design, Power Electronics, Microcontroller & ADC Integration (ESP32-S3, ADS1299), Mechanical CAD (Onshape)
- **Programming:** C++, Python, Git, PlatformIO
- **Concepts:** Embedded Firmware, Real-Time Systems (RTXI, Xenomai), Protocols (SPI, I2C, UART, TCP/IP, LSL), Biosignal Processing, Closed-Loop Control, Neural Interface Hardware, Hardware Fault Protection